

LAYER 1: TACTICAL COMMUNICATION LINK LAYER (MIL-STD PROFILES)

- VLF (Submarine): 300 bps, $T_{\text{budget}}=30\text{s}$
- HF (Ground): 1.2 kbps, $T_{\text{budget}}=5\text{s}$
- UHF (Vehicle): 64 kbps, $T_{\text{budget}}=0.5\text{s}$
- Ku-SATCOM: 2 Mbps, $T_{\text{budget}}=0.8\text{s}$

↓ Link State Vector $x = [\text{BW}, \text{Eb/N0}, \text{BER}, \text{CPU}, \text{Threat}, \text{Queue}, \text{Overhead}, \text{Margin}]$

LAYER 2: FEASIBILITY FILTER & LinUCB CONTEXTUAL BANDIT SELECTOR

- Hard Latency Constraint Filter: $T_{\text{total}} = T_{\text{comp}} + T_{\text{trans}} \leq T_{\text{budget}}$
- LinUCB Selector ($d=8$): $p_{\{a,t\}} = \theta_a^T x + \alpha \sqrt{(x^T A_a^{-1} \{-1\} x)}$
- Threat-Weighted Reward: $r = \lambda_1 \cdot \text{Security} + \lambda_2 \cdot (1 - \text{Latency}) + \lambda_3 \cdot (1 - \text{BW})$

↓ Selected PQC Pair & Weight Vectors θ_a^*

LAYER 3: DUAL-GRANULARITY XAI EXPLANATION ENGINE

- Operator Summary: Top-k feature attributions ($\theta_a^* \cdot x$) & natural language rationale
- Auditor Trace: Full attribution JSON audit log ($< 10 \mu\text{s}$ explanation time)
- Commander Override Interface: Human-in-the-loop trust management

↓ Feature Attribution Vectors & Transition Request

LAYER 4: ADVERSARIAL CHANNEL-SPOOFING DETECTOR (NOVELTY 4)

- Signal 1 (Physical BER Consistency): $P_b = \frac{1}{2} \text{erfc}(\sqrt{E_b/N_0})$ verification
- Signal 2 (XAI Contribution Anomaly): Clean baseline z-score deviation monitoring
- Alert Engine: CRITICAL (Block Downgrade) | WARNING (Flag Review) | CLEAR